

Drayan Mathematical Series

Begginer Olympiad Algebra

Problem

Let $P(x) = x^{2025}$ and let $R(x)$ denote the remainder when $P(x)$ is divided by

$$Q(x) = x^4 + x^3 + 2x^2 + x + 1.$$

If

$$P(x) = Q(x)S(x) + R(x),$$

where $\deg R < 4$, determine $R(3)$.

IOQM 2025 — Problem 24

Solution

We first factor the divisor:

$$Q(x) = x^4 + x^3 + 2x^2 + x + 1.$$

Grouping terms appropriately,

$$Q(x) = (x^4 + x^3 + x^2) + (x^2 + x + 1) = x^2(x^2 + x + 1) + (x^2 + x + 1),$$

whence

$$Q(x) = (x^2 + 1)(x^2 + x + 1).$$

Now, modulo $x^2 + 1$, we have

$$x^2 \equiv -1,$$

which immediately gives $x^4 \equiv 1$. Similarly,

$$x^3 - 1 = (x - 1)(x^2 + x + 1),$$

so that $x^3 \equiv 1 \pmod{x^2 + x + 1}$.

Since $\text{lcm}(4, 3) = 12$, it follows that

$$x^{12} \equiv 1 \pmod{x^2 + 1} \quad \text{and} \quad x^{12} \equiv 1 \pmod{x^2 + x + 1}.$$

Because the factors $x^2 + 1$ and $x^2 + x + 1$ are relatively prime, we conclude that

$$x^{12} \equiv 1 \pmod{Q(x)}.$$

Using $2025 = 12 \cdot 168 + 9$, we obtain

$$x^{2025} = (x^{12})^{168} x^9 \equiv x^9 \pmod{Q(x)}.$$

Thus it suffices to determine the remainder of x^9 upon division by $Q(x)$. Let this remainder be $R(x)$.

Since

$$x^9 = (x^3)^3 \equiv 1 \pmod{x^2 + x + 1},$$

we have $R(x) \equiv 1 \pmod{x^2 + x + 1}$. Further,

$$x^9 = x(x^4)^2 \equiv x \pmod{x^2 + 1},$$

and therefore $R(x) \equiv x \pmod{x^2 + 1}$.

As $\deg R < 4$, the first congruence implies that

$$R(x) = (ax + b)(x^2 + x + 1) + 1$$

for suitable constants a and b . Expanding,

$$R(x) = ax^3 + (a + b)x^2 + (a + b)x + (b + 1).$$

Reducing modulo $x^2 + 1$, where $x^2 \equiv -1$ and $x^3 \equiv -x$, gives

$$R(x) \equiv -ax - (a + b) + (a + b)x + (b + 1) = bx + (1 - a).$$

Comparing with $R(x) \equiv x \pmod{x^2 + 1}$, we obtain

$$b = 1, \quad a = 1.$$

Hence

$$R(x) = (x + 1)(x^2 + x + 1) + 1 = x^3 + 2x^2 + 2x + 2.$$

Finally,

$$R(3) = 3^3 + 2 \cdot 3^2 + 2 \cdot 3 + 2 = 27 + 18 + 6 + 2 = 53.$$

Answer

$$R(3) = 53$$